Naive Bayes Classifier | Part 6 | Intuition

1 The Big Picture

Naive Bayes is a **probabilistic classifier** based on **Bayes' Theorem**.

It predicts the class of a data point by answering:

👉 *"Given the features, what is the probability of belonging to each class?"*

Formally:

$$P(Class|Features) = \frac{P(Features|Class) \cdot P(Class)}{P(Features)}$$

Then, it picks the class with the **highest probability**.

2 The "Naive" Assumption

Why is it called **Naive**?

- Because it assumes that **all features are independent given the class**.
- In real life, this is often **not true** — but surprisingly, the algorithm still works very well.

👉 ***Example:*** For email spam detection

- Features = {"Offer", "Discount", "Click"}.
- Naive Bayes assumes these words occur independently in spam emails, which isn't strictly correct (they often co-occur).
- But this assumption makes computation simple and still gives accurate results.

3 How the Logic Works (Step by Step)

1. **Start with Prior**: The probability of each class (e.g., Spam = 40%, Not Spam = 60%).
2. **Multiply by Likelihoods**: For each feature, calculate how likely it is to appear in that class.

$$P(Spam|Words) \propto P(Spam) \cdot P(Word_1|Spam) \cdot P(Word_2|Spam) \cdot \ldots$$

3. **Normalize**: Compare probabilities across all classes and pick the class with the highest.

4 Intuitive Example

Imagine you want to predict whether someone likes **Pizza** or **Salad** based on their choices.

- Priors:
    1. P(LikesPizza) = 0.6

2. P(LikesSalad) = 0.4

- Likelihoods:
  1. If Pizza → more likely to like Cheese, Pepperoni
  2. If Salad → more likely to like Lettuce, Cucumber

Now if a new person likes **Cheese + Pepperoni**, Naive Bayes will strongly lean toward **Pizza** because those features fit Pizza much more than Salad.

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5 Why It Works Well

- Even if the independence assumption is not strictly true, probabilities still often **rank the correct class higher**.
- Works especially well with **high-dimensional data** like text (many words/features).
- Simple, fast, and effective for baseline models.

✓ Summary:

Naive Bayes = Bayes' Theorem + Independence Assumption.

It calculates probabilities for each class, multiplies feature likelihoods, and chooses the class with the highest probability.
